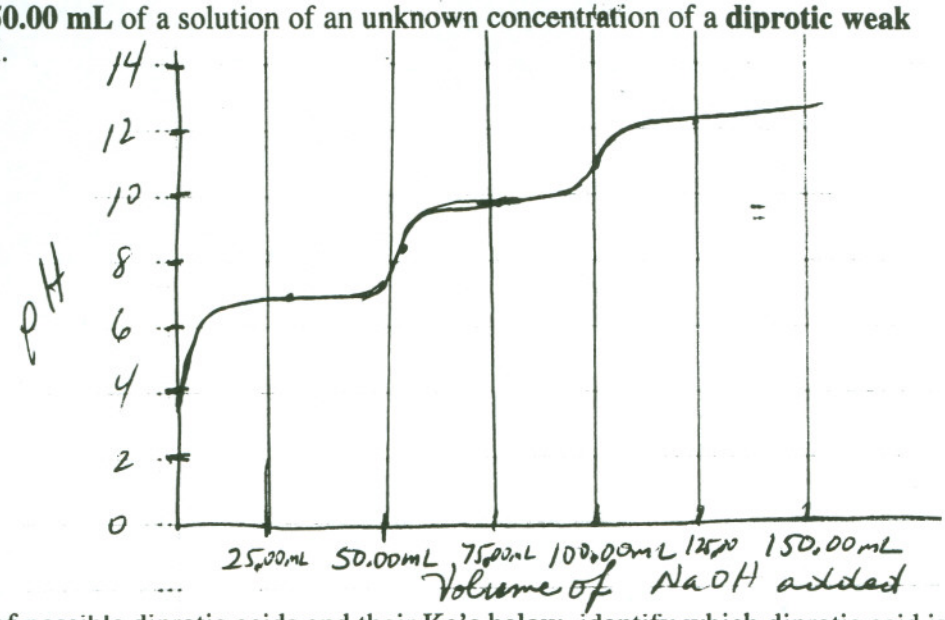
Analytical Cume (Dec. 3, 1999)
Theme: Acids and Bases and Neutralization

1. (40 points) Extracting information from a titration curve of a diprotic acid.

Below is a titration curve of **50.00 mL** of a solution of an unknown concentration of a **diprotic weak acid** with **0.1000 M NaOH**.



(a) (25 points) Given the list of possible diprotic acids and their Ka's below, identify which diprotic acid is being titrated and explain how you arrived at your decision.

Acid, Formula	Ka_1	Ka_2
ethylene diammonium ion, $+H_3NCH_2CH_2NH_3+$	1.42×10^{-7}	1.18×10^{-10}
fumaric acid, trans-HOOCCH:CHCOOH	8.85×10^{-4}	3.21×10^{-5}
hydrogen sulfide, H_2S	9.6×10^{-8}	1.3×10^{-14}
maleic acid, cis-HOOCCH:CHCOOH	1.3×10^{-2}	5.9×10^{-7}

(b) (15 points) What is the starting concentration of the diprotic acid? Show work or explain.

2. (30 points) Application of a Neutralization Titration

The Kjeldahl method is the most common method for determining organic nitrogen (such as percent nitrogen, %N) and is based upon a neutralization titration. Since most proteins contain approximately the same percentage of nitrogen, multiplication of %N by a suitable factor (6.25 for meats, 6.38 for dairy products, and 5.70 for cereals and grains) gives the percentage of protein in a sample.

A **0.7121 g** sample of a wheat flour was analyzed by the Kjeldahl method. The ammonia formed by addition of concentrated base after digestion with H_2SO_4 was distilled into **25.00 mL** of **0.04977 M** HCl. The excess HCl was then back-titrated with **3.97 mL** of **0.04012 M NaOH**. Calculate the percent protein in the flour.

3. (30 points) Acid/Base Indicator Dyes

When choosing an indicator to show when the equivalence point of an acid/base titration is achieved, one picks an indicator dye which changes color near the equivalence point. Typically, the color change of the indicator occurs when the pH of the titration solution changes in the range

$pK_{HIn} \pm 1$,

where $pK_{HIn} = -\log(K_{HIn})$,

and K_{HIn} is the acid dissociation constant for the indicator. Thus, one tries to match the pH of the equivalence point in the titration to the middle of the $pK_{HIn} \pm 1$ range of the indicator.

(a) (20 points) Given that an acid/base indicator has the following equilibrium (HIn is the acid form of the indicator with the color purple and In- is the base form of the indicator with the color yellow):

$$\mathbf{HIn + H_2O \Longleftrightarrow In^- + H_3O^+}$$

which has an acid dissociation equilibrium constant of $\mathbf{K_{HIn}}$;

and given that we can SEE the acid color when

$$\mathbf{[HIn] / [In^-] = 10}$$

and we can SEE the base color when

$$\mathbf{[HIn] / [In^-] = 1/10},$$

(i) Derive an equation that relates the **pH** of the solution to the $\mathbf{pK_{HIn}}$, at the point when you expect to SEE the **acid** color.

(ii) Derive an equation that relates the **pH** of the solution to the $\mathbf{pK_{HIn}}$, at the point when you expect to SEE the **base** color.

(iii) Show that the pH transition range for an indicator color change is $\mathbf{pH = pK_{HIn} \pm 1}$.

(b) (10 points) Identify the pH of the second equivalence point in the titration curve in Problem 1 above. Which of the following indicators would be the best candidate to indicate when the endpoint is reached? Explain why.

Common Name	pK_{HIn}	Color Change
Thymol blue	1.65	red to yellow
Bromothymol blue	7.10	yellow to blue
Cresol purple	8.4	yellow to purple
Thymolphthalein	9.9	colorless to blue
Alizarin yellow GG	11	colorless to yellow